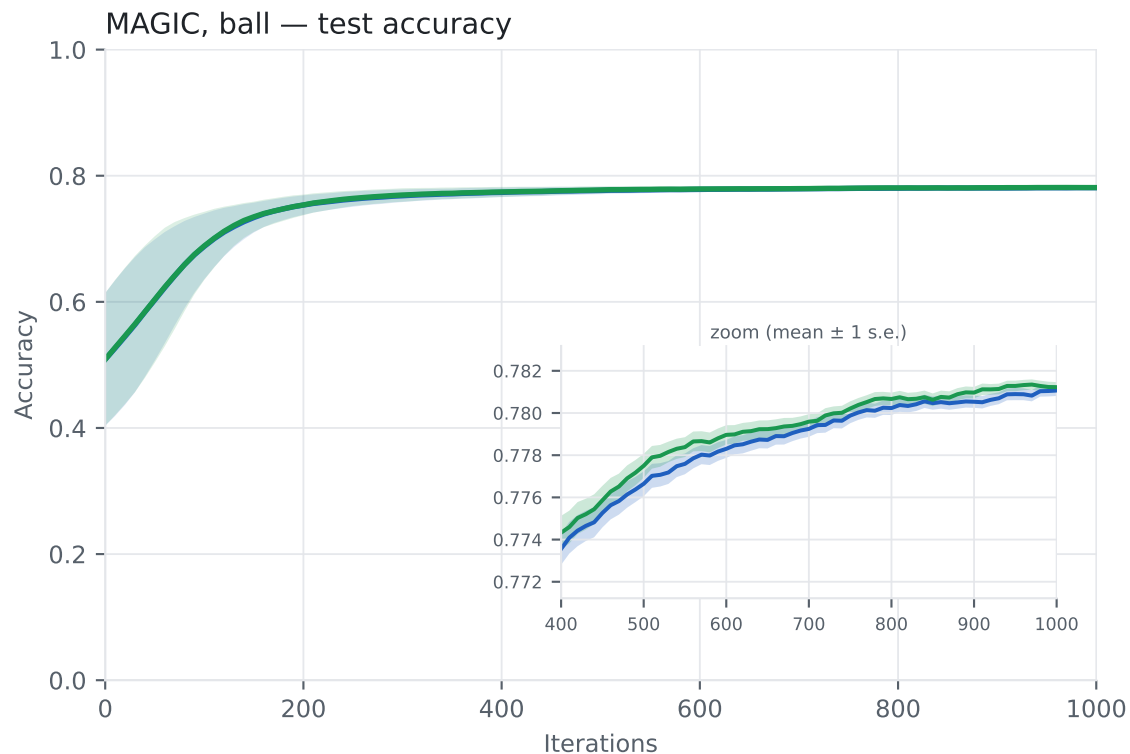
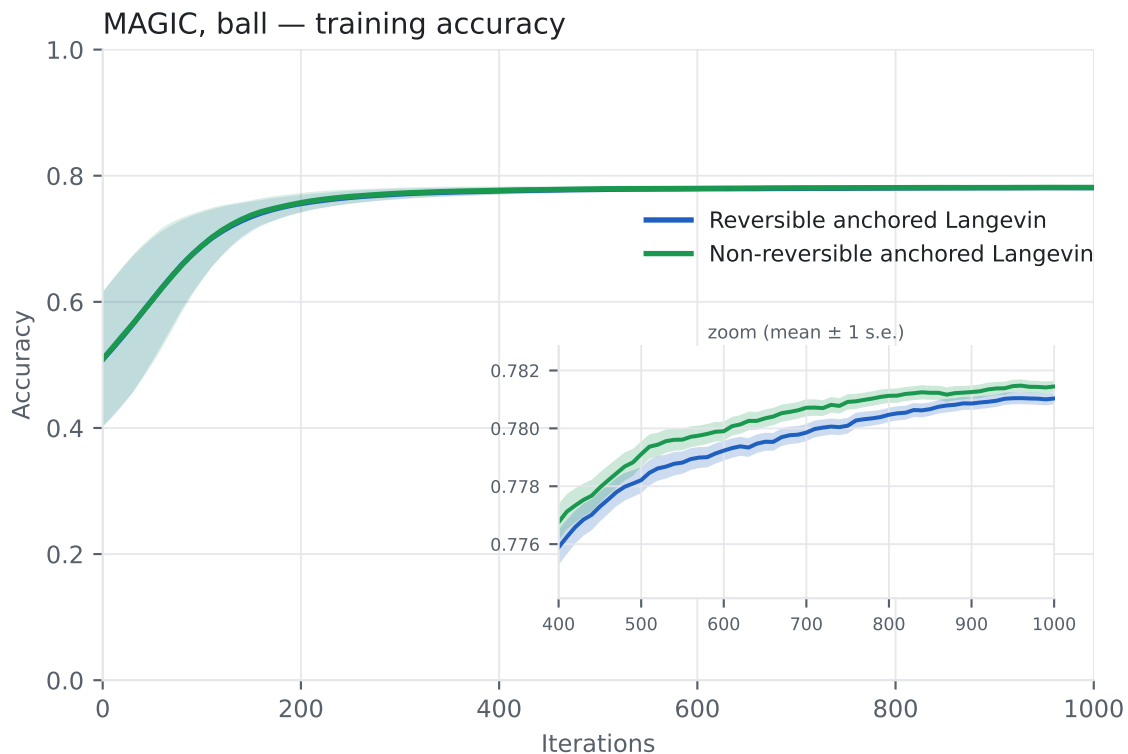


magic_ball: exact gradient, LASSO anchor



MAGIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + nine features), no mini-batching: the EXACT gradient is used at every step. $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j=1}^9 |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. $\sigma = 10$, $\lambda = 152.2 = 0.01 n_{\text{train}}$, $\delta = 0.0005062$, so a in $[0.50, 1]$. Block strengths $s = (5, 5, 5)$ on $w_{1..9}$ only. Step $\eta = 1e-06$. $R = 100$ coupled replicates (same W_0 , same Gaussian stream) on one fixed stratified 80/20 split ($n_{\text{train}} = 15216$, $n_{\text{test}} = 3804$). Paired final-iterate difference, non-reversible minus reversible: training $+0.0004$ ($t = +2.35$), test $+0.0002$ ($t = +0.65$). Bands: mean $\pm$ 1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval.